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| **Fiscal Year:** | FY 2017 | **Task Last Updated:** | FY 10/12/2017 |
| **PI Name:** | Paul, Anna-Lisa Ph.D. | | |
| **Project Title:** | Epigenetic Change in Arabidopsis thaliana in Response to Spaceflight - Differential Cytosine DNA Methylation of Plants on the ISS | | |
| **Division Name:** | Space Biology | | |
| **Program/Discipline:** | | | |
| **Program/Discipline--Element/Subdiscipline:** | SPACE BIOLOGY--Cellular and molecular biology | | |
| **Joint Agency Name:** | | **TechPort:** | No |
| **Human Research Program Elements:** | None | | |
| **Human Research Program Risks:** | None | | |
| **Space Biology Element:** | (1) Cell & Molecular Biology<br>(2) Plant Biology | | |
| **Space Biology Cross-Element Discipline:** | (1) Reproductive Biology<br>(2) Developmental Biology | | |
| **Space Biology Special Category:** | None | | |
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| **Zip Code:** | 32611-0690 | **Congressional District:** | 3 |
| **Comments:** | | | |
| **Project Type:** | Flight | **Solicitation / Funding Source:** | 2014 Space Biology Flight NNH14ZTT001N |
| **Start Date:** | 09/12/2014 | **End Date:** | 09/10/2018 |
| **No. of Post Docs:** | 2 | **No. of PhD Degrees:** | 1 |
| **No. of PhD Candidates:** | 2 | **No. of Master' Degrees:** | 1 |
| **No. of Master's Candidates:** | 0 | **No. of Bachelor's Degrees:** | |
| **No. of Bachelor's Candidates:** | 2 | **Monitoring Center:** | NASA KSC |
| **Contact Monitor:** | Levine, Howard | **Contact Phone:** | 321-861-3502 |
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| **Flight Program:** | ISS | | |
| **Flight Assignment:** | ISS<br>NOTE: Extended to 9/10/2018 per NSSC information (Ed., 10/10/17) | | |
| **Key Personnel Changes/Previous PI:** | | | |
| **COI Name (Institution):** | Ferl, Robert Ph.D. ( University of Florida ) | | |
| **Grant/Contract No.:** | NNX14AT24G | | |
| **Performance Goal No.:** | | | |
| **Performance Goal Text:** | | | |

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| **Task Description:** | This project seeks to use an unbiased genome-wide approach to define the changes in DNA methylation that occur in Arabidopsis during spaceflight, in order to better understand the spectrum of responses that allow adaptation to the spaceflight environment. This project is one of both discovery on a genome-wide scale and application to specific hypotheses. We propose replicated whole genome bisulfite sequencing in Arabidopsis roots and leaves to discover and quantify 5-methylCytosine (5mCyt) epigenetic modification, coupled with RNASeq examination of the comparable transcriptomes to both quantitatively and qualitatively assess spaceflight-associated DNA methylation on transcript profiles. We will then test the hypothesis that methylation is involved in spaceflight adaptation by similarly examining methylation mutants.<br>Epigenetic features play a role in the regulation of gene expression and the subsequent response of an organism to its environment. DNA methylation is a major epigenetic modification that is not directly coded into the genome sequence and yet can modify expression and be inherited for at least for one generation. Several groups, including ours, have shown that large numbers of plant genes are differentially expressed in response to spaceflight.<br><br>We now have extensive preliminary data that show significant genome-wide 5mCyt modification differences in spaceflight plants compared to ground controls.<br><br>The tactical objectives of this proposal are: 1) Bisulfite whole genome sequencing will be used to generate maps of spaceflight-induced changes in 5mCyt at single base-pair resolution for roots and leaves. 2) RNASeq will be used to digitally quantify and qualitatively assess transcriptome in the same samples, as differential methylation has been functionally linked to changes in both promoter regulation and structural changes in RNAs. These data will characterize functional impacts of spaceflight induced across the entire genome. 3) DNA methylation mutants will be used to directly test the hypothesis that epigenetic modification of DNA is a key aspect of spaceflight adaptation.<br><br>Elucidating the spaceflight methylomes of plants contributes to the fundamental understanding of how biology responds and adapts to the spaceflight environment, and also provides insight in the adaptive strategies plants use in the face of novel stimuli outside of their evolutionary experience. The long term goal of this line of inquiry is to understand how the methylome contributes to the adaptive strategies of plants to novel environments. |
| **Rationale for HRP Directed Research:** | |
| **Research Impact/Earth Benefits:** | This research contributes to a greater understanding of the role of epigenetic changes to the genome in response to novel or stressful environments. |
| **Task Progress:** | In this year, the APEX04 experiment was launched.<br>However, before the flight experiment was launched, the biochemical protocols were first calibrated with non-spaceflight experiment samples:<br><br>The EVT (Experiment Verification Test) materials were used to calibrate the amount of material needed to provide the 200 ng of genomic DNA required per sample to conduct the methylome analyses. The results of these calibrations were published in December of 2016 (LeFrois et al., 2016). Briefly: Genomic DNA was isolated from the three genotypes comprising APEX04: Col-0, met1-7, and elp2-5. Molecular analysis was performed to assess both the quality and quantity of gDNA produced from these extractions. Gene-specific DNA methylation (rather than global, due to cost constraints) was evaluated at two sites: AT2G07698 (+253 to +507 from ATG) and AT2G07687 (+19 to +355 from ATG) from shoot DNA samples of Col-0, met1-7 and elp2-5.<br><br>For the launch, the preparation began with the transport and turnover of the plates. A total of 30 plates each day were prepared and planted in the Principal Investigator (PI) laboratory on February 11 & February 13, 2017. Wrapped plates were then transported to Kennedy Space Center (KSC) for turnover on February 15, 2017. They were transported in a cooler to and kept as close to 4 degrees as possible during travel. A ground control was also prepared and installed into the Veggie unit housed in the International Space Station Environmental Simulator chamber (ISSES). An additional experiment of opportunity substantially enhanced the imaging objectives of the APEX-04 experiment. One of the Col-0 spares was replaced with Col-0 plants engineered with GFP and RFP reporter genes.<br><br>On orbit and GC operations:<br><br>Imaging – SLR camera: Images of the first plates in VEGGIE (A1, B1, and C1) were taken on Day 4, Day 8, and Day 11(harvest). Due to excellent germination and growth all plates appear to have consistent growth. Photography of plates was sufficient for post-harvest analysis.<br><br>Harvests: Plants for SVT and EVT were grown in VEGGIE and were harvested at 11 days. Each plate was harvested to a single Falcon tube containing RNA later. Samples were then stowed at -80Deg Celsius until returned to PI's laboratory. Harvest from station was monitored by the PI via video communication from their lab in Gainesville, FL.<br><br>Post-Harvest Analysis --Samples for both flight and ground control were transferred by the PI to the University of Florida for analysis of quality and quantity. Samples were warmed and separated into root and leaf tissue; hypocotyl tissue was removed and stored for future analysis. Plates were then separated into 4 replicates, each having two plates within a replicate. A portion of the plants (2-4) were removed from each plate for RNA extraction, and then depending on genotype, 4-20 plants that were left on the plate were used for DNA extraction.<br><br>Extraction of Nucleic Acids from Flight and Ground Control Samples--RNA extraction was completed using a Qiagen RNeasy Plant kit. DNA extraction was preformed using a method developed in-house for high levels of clean genomic DNA. All sample material provided enough nucleic acid material for downstream applications, and those analyses are currently underway. |
| **Bibliography Type:** | Description: (Last Updated: 11/20/2025) |
| **Articles in Peer-reviewed Journals** | LeFrois CE, Zhou M, Moraga Amador D, Sng N, Paul A-L, Ferl RJ. "Enabling the spaceflight methylome: DNA isolated from plant tissues preserved in RNAlater® is suitable for bisulfite PCR assay of genome methylation." Gravit Space Res. 2016 Dec;4(2):28-37. http://gravitationalandspacebiology.org/index.php/journal/article/view/745 , Dec-2016 |